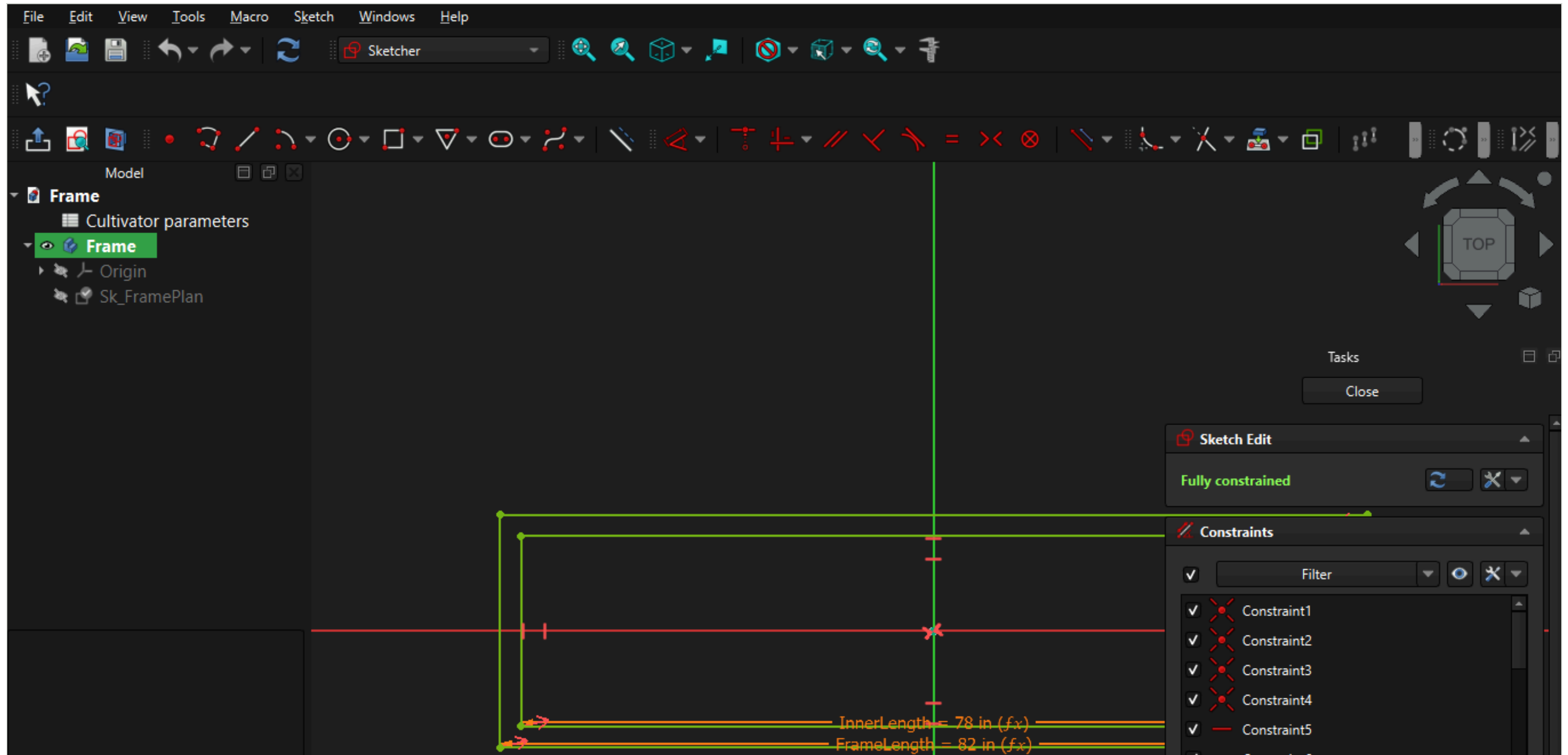
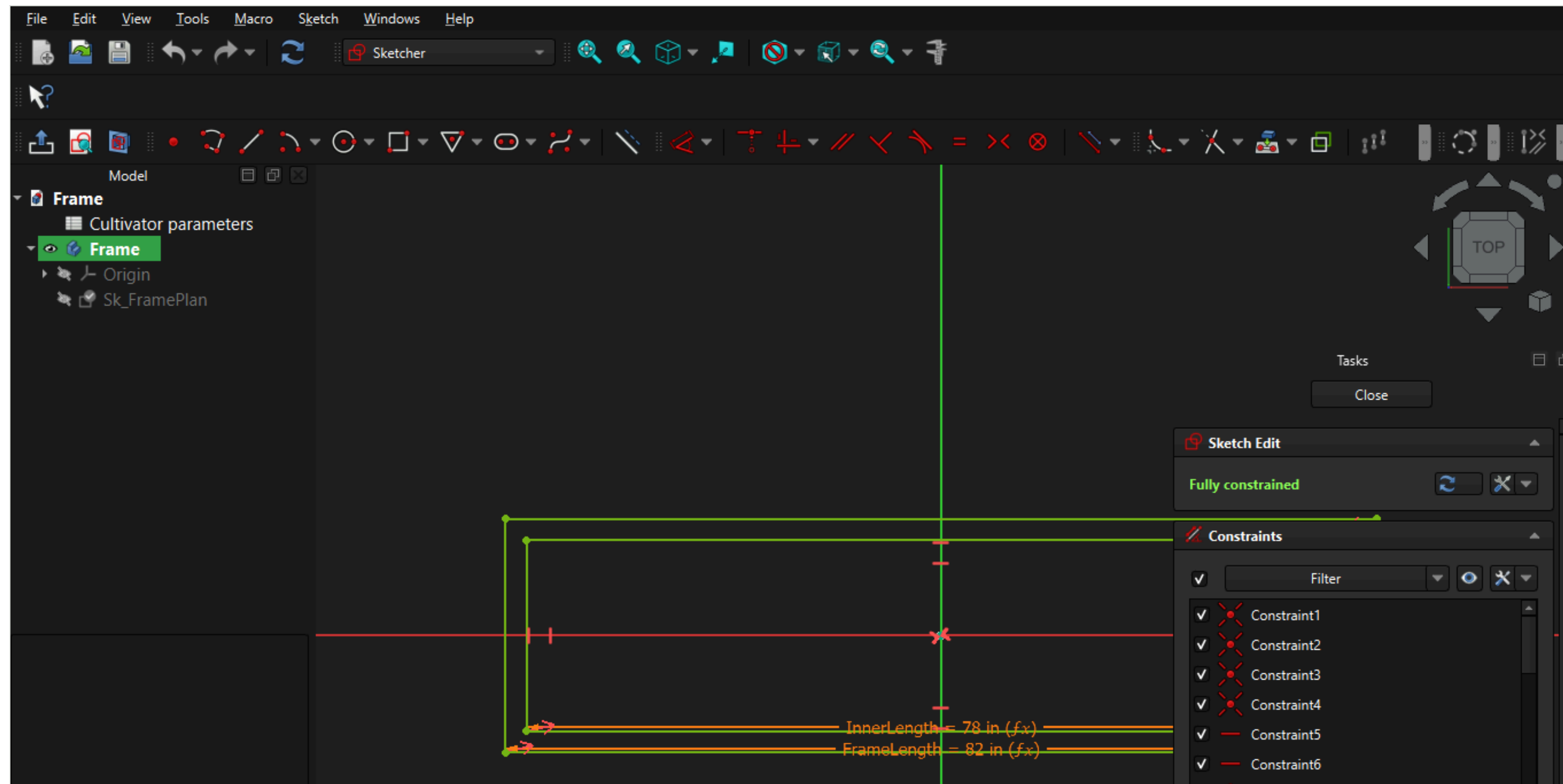


Sketching and Constraints

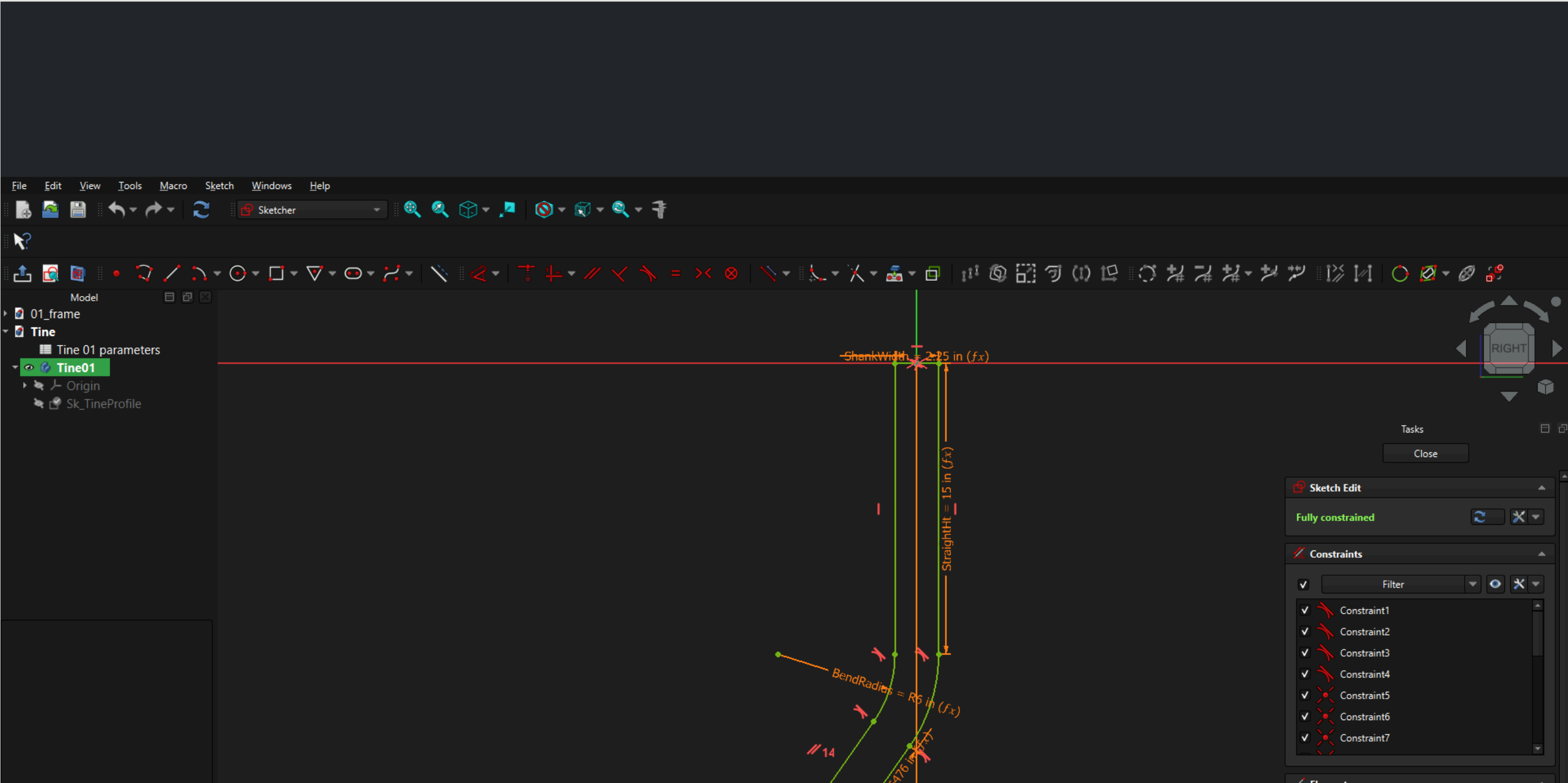


What fully constrained means



DO NOT LEAVE IT FOR LATER

Why it matters



The two kinds of constraint

RULES ABOUT RELATIONSHIPS

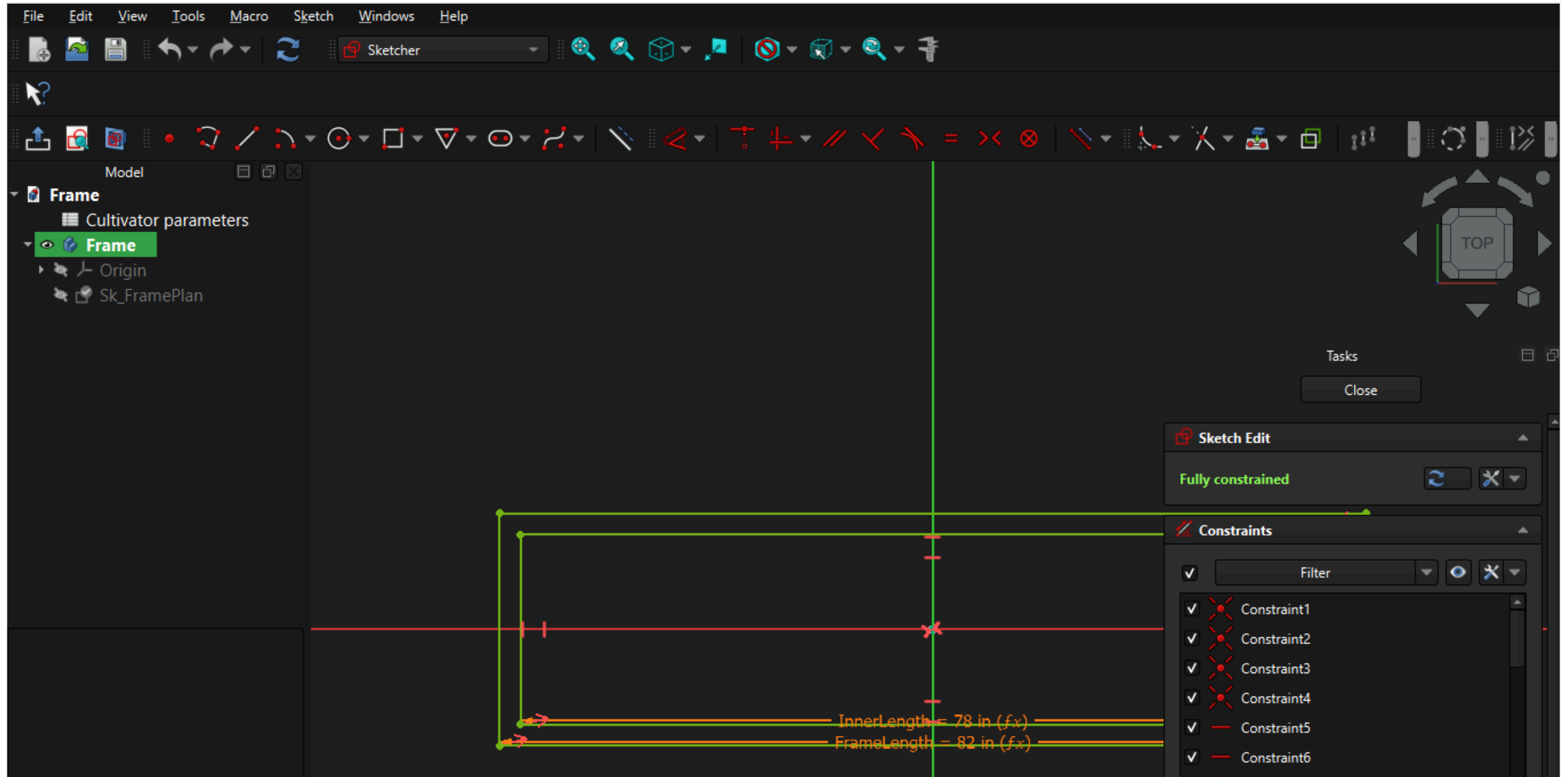
RULE	KEY	IT SAYS
Coincident	C	these two points are the same point
Horizontal	H	this line is level
Vertical	V	this line is upright
Parallel	P	these two lines run the same way
Tangent	T	the curve meets the line smoothly
Equal	E	same length, or same radius
Symmetric	S	these mirror about that line

RULES WITH NUMBERS

RULE	KEY	IT SAYS
Distance	K D	this is 15 inches long
Horizontal distance	L	these are 3 inches apart sideways
Vertical distance	I	these are 2 inches apart up and down
Radius	K R	this arc is R6
Diameter	K O	this hole is half an inch

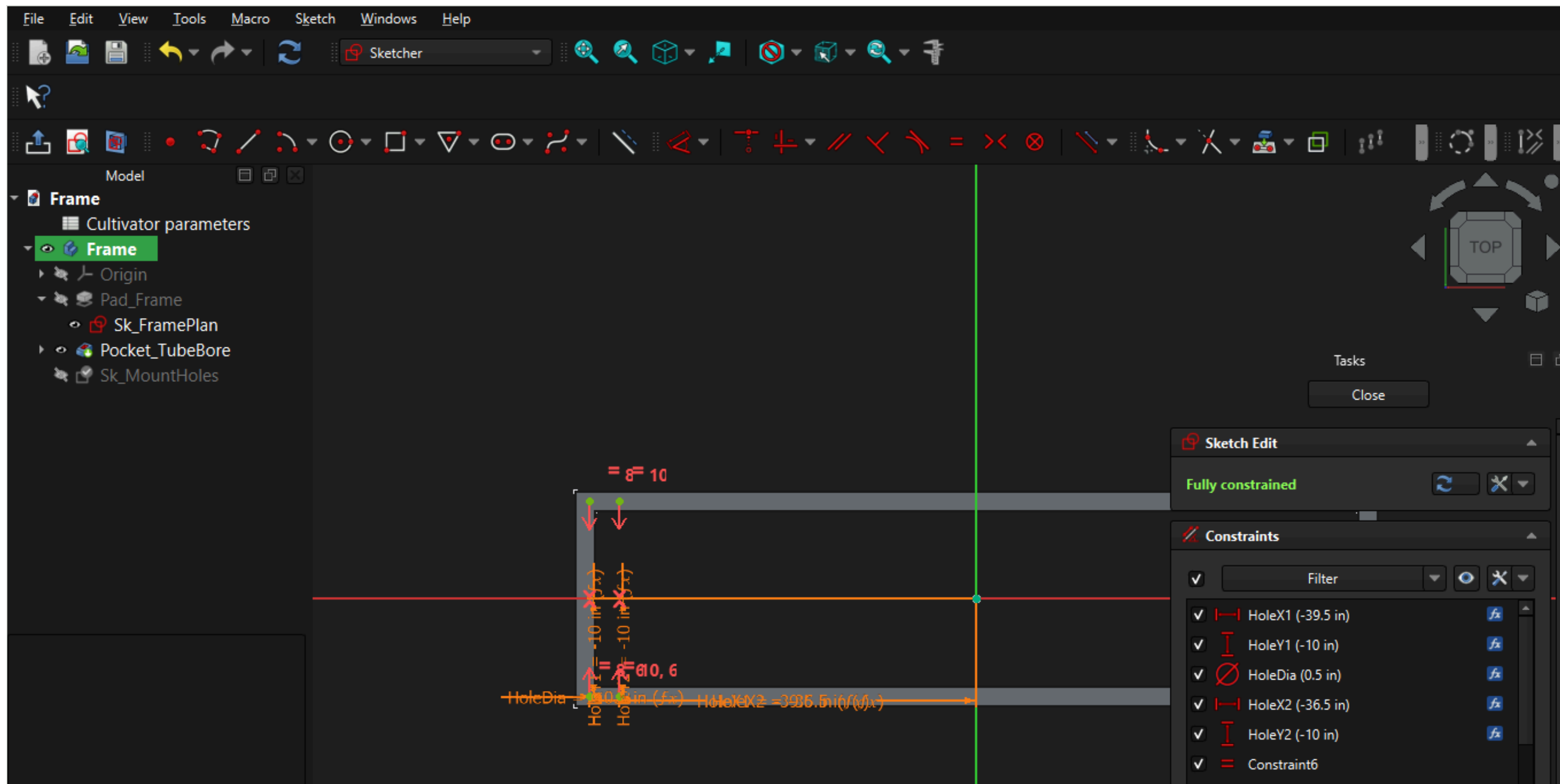
Prefer the first kind. Two lines told to be **equal** stay equal forever. Two lines each measured 5 inches stay equal only until somebody edits one.

A centred rectangle, step by step

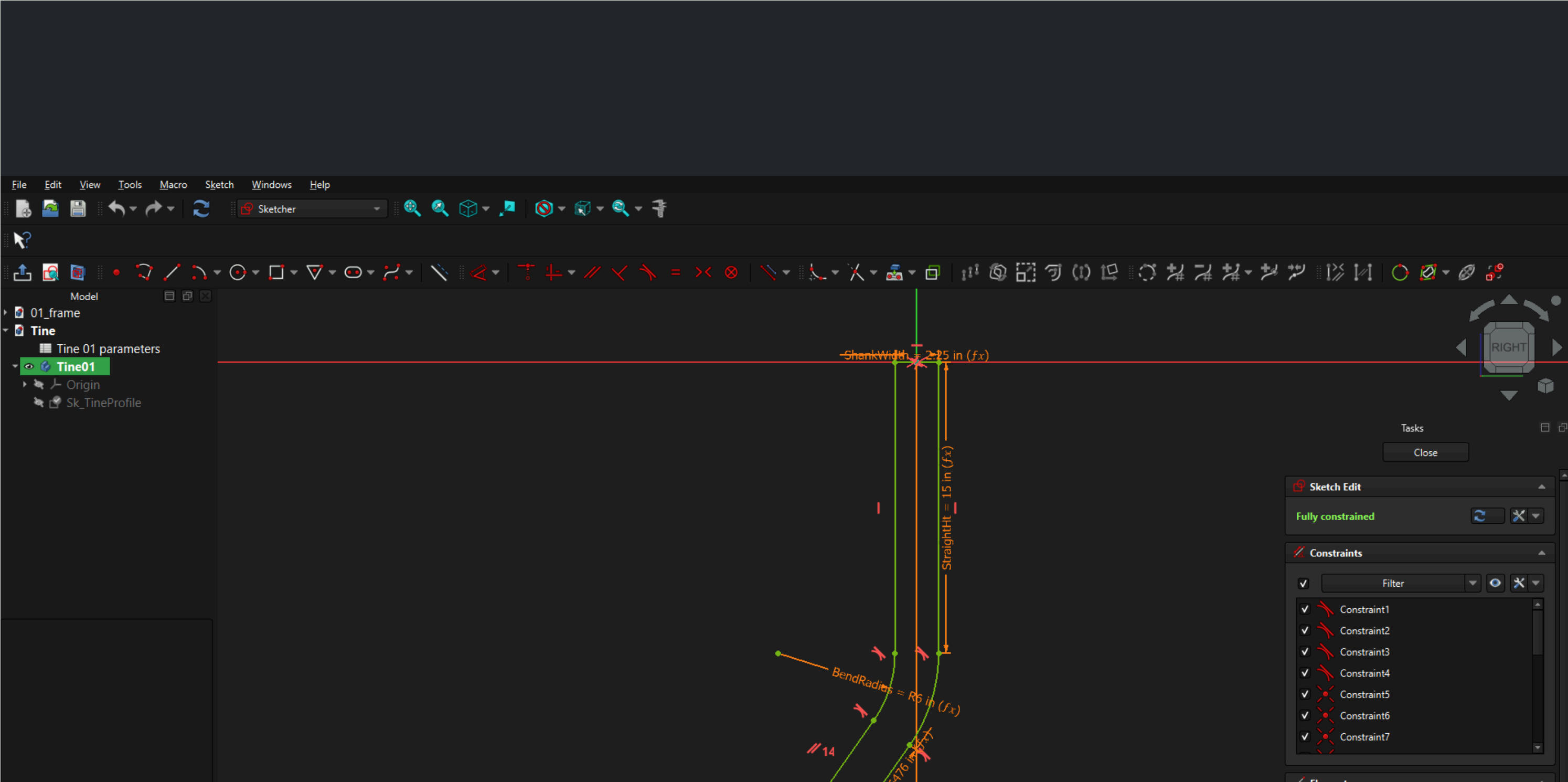


ONE OF THE MOST USEFUL RULES THERE IS

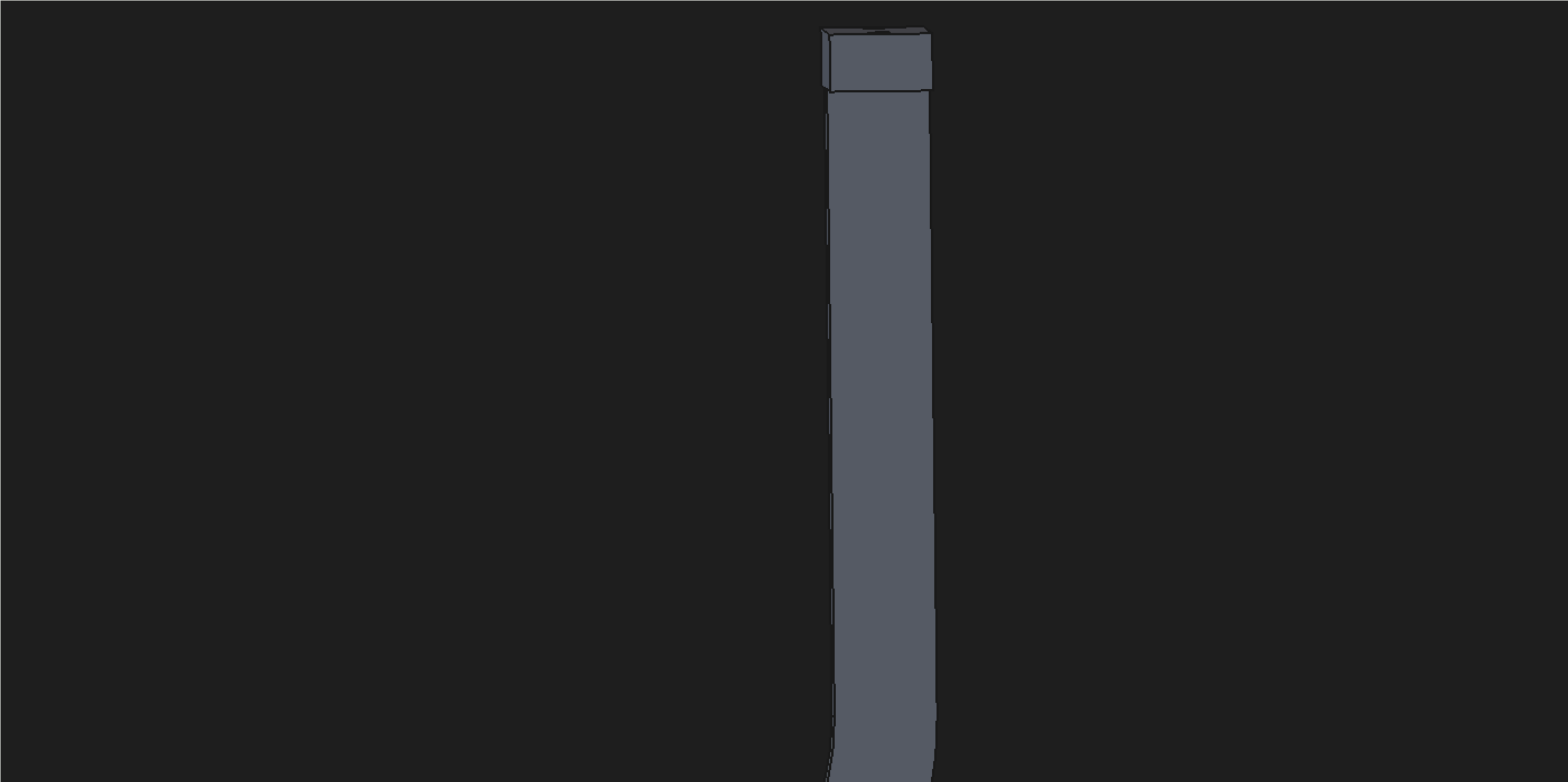
Symmetry beats copying



Trap one: joining a line to a curve



Trap two: angles have two answers



Sketching in the right order

WORK THIS WAY

- 1 **Draw roughly.** Do not chase exact positions with the mouse. Get the shape right.
- 2 **Add the relationship rules.** Horizontal, vertical, tangent, equal, symmetric.
- 3 **Then add the numbers,** watching the count go down.
- 4 **Stop at zero.** Do not add one more for luck — that is the one that causes trouble.

TWO HABITS WORTH FORMING

- **Name your dimensions.** Double-click the constraint and call it `FrameLength`, not `Constraint7`. You need the name to drive it from a spreadsheet later.
- **Close your shapes.** An open outline cannot be padded. If a Pad complains, look for a gap at a corner.

If a Pad or Pocket fails and the sketch looks perfectly fine, read the small message in the Sketcher panel first. It is usually telling you exactly what is wrong.